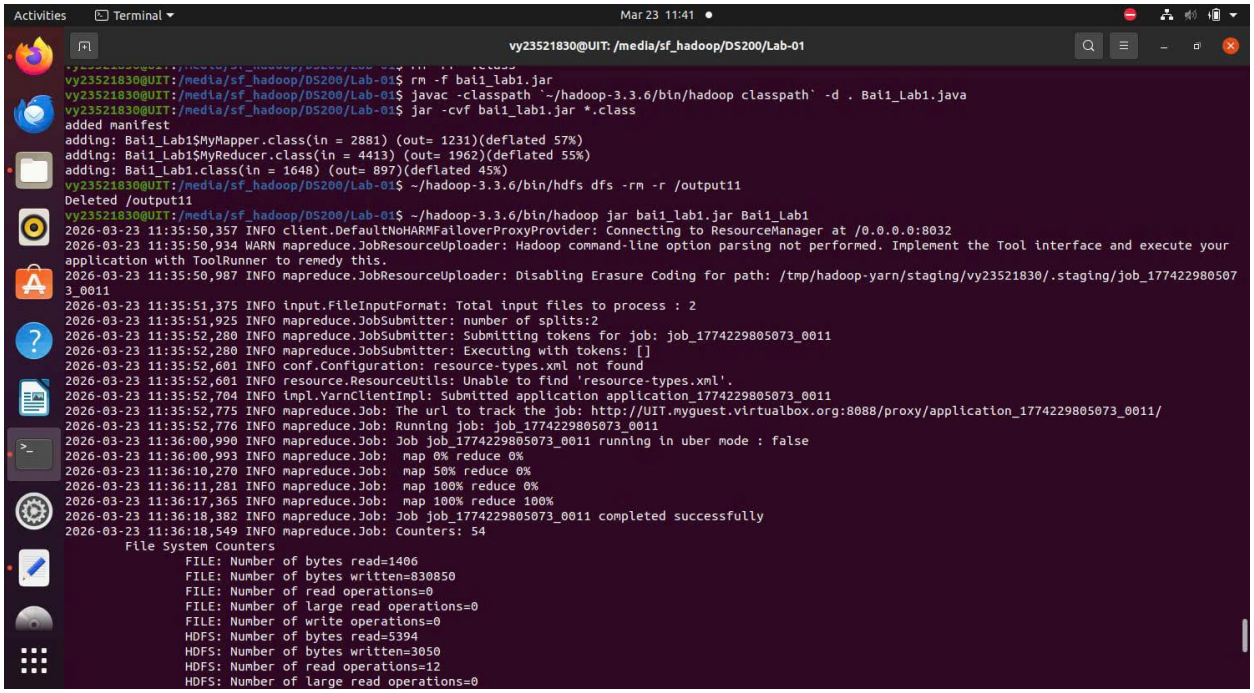
MSSV: 23521830

Họ và tên: Nguyễn Hoàng Vy

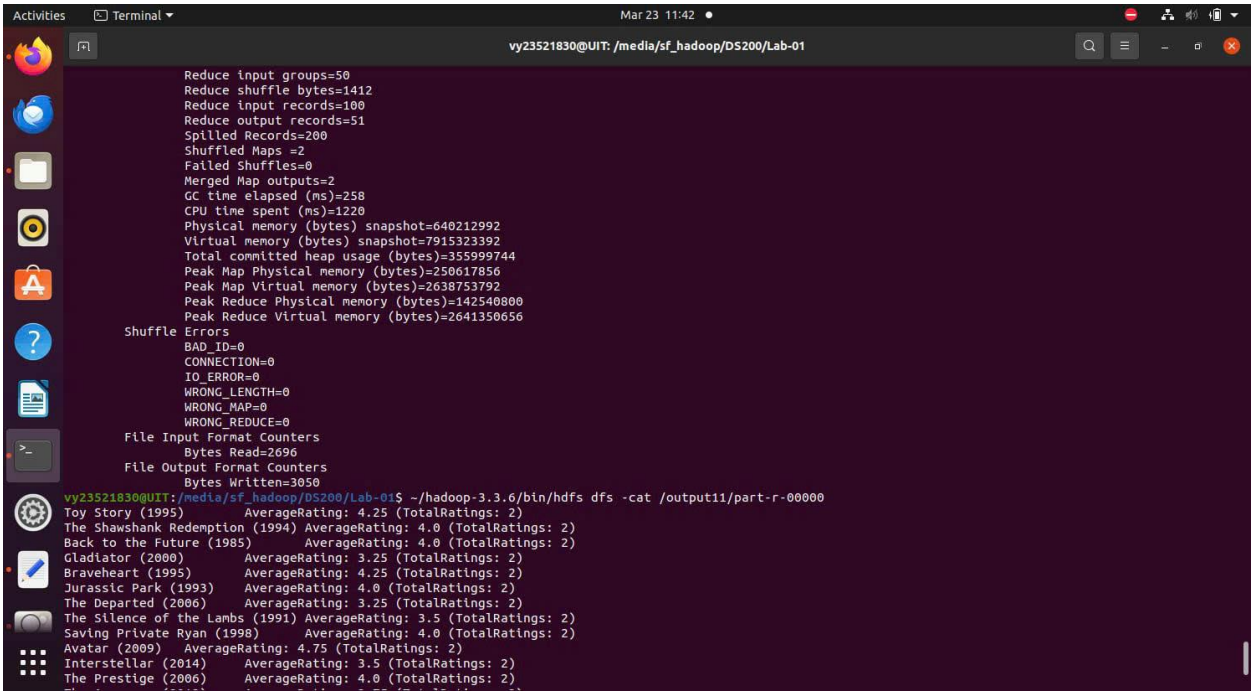
Lớp: DS200.Q21

Bài tập Thực hành 1

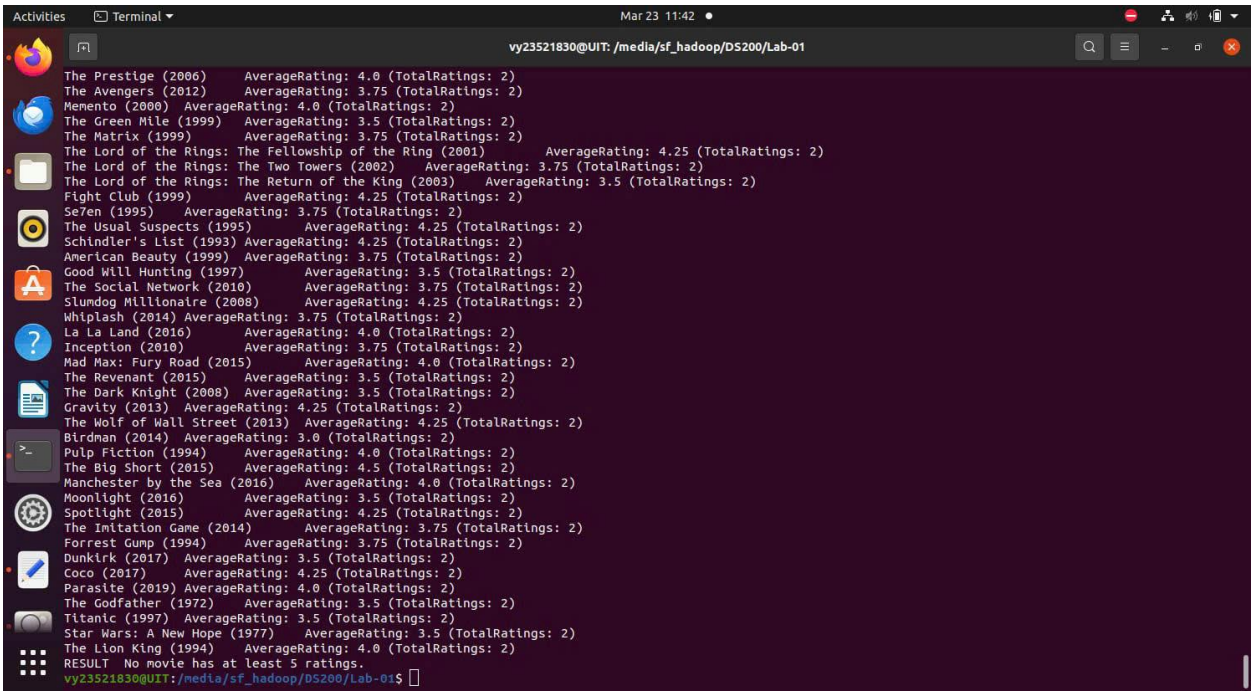
Câu 1:



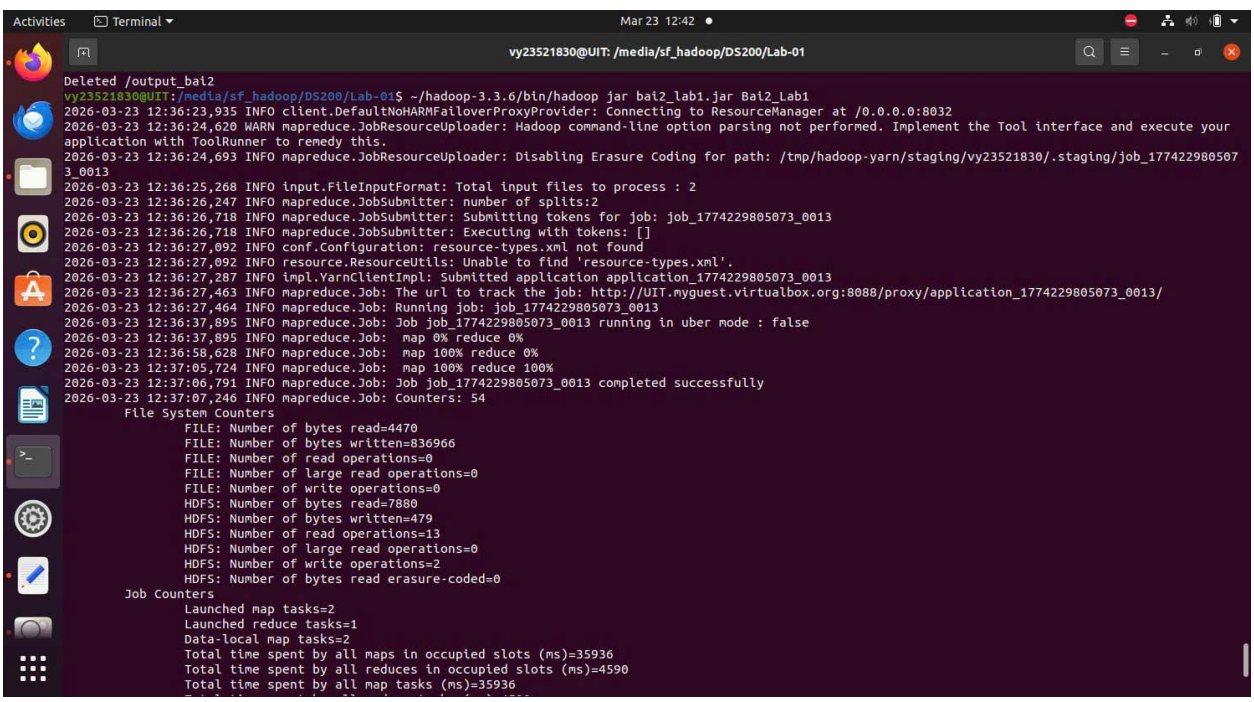
```
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ rm -f bai1_lab1.jar
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ javac -classpath `~/hadoop-3.3.6/bin/hadoop classpath` -d . Bai1_Lab1.java
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ jar -cvf bai1_lab1.jar *.class
added manifest
adding: Bai1_Lab1$MyMapper.class(in = 2881) (out= 1231)(deflated 57%)
adding: Bai1_Lab1$MyReducer.class(in = 4413) (out= 1962)(deflated 55%)
adding: Bai1_Lab1.class(in = 1648) (out= 897)(deflated 45%)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hdfs dfs -rm -r /output11
Deleted /output11
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hadoop jar bai1_lab1.jar Bai1_Lab1
2026-03-23 11:35:50,357 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-23 11:35:50,934 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your
application with ToolRunner to remedy this.
2026-03-23 11:35:50,987 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/vy23521830/.staging/job_177422980507
3_0011
2026-03-23 11:35:51,375 INFO input.FileInputFormat: Total input files to process : 2
2026-03-23 11:35:51,925 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-23 11:35:52,280 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774229805073_0011
2026-03-23 11:35:52,280 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-23 11:35:52,601 INFO conf.Configuration: resource-types.xml not found
2026-03-23 11:35:52,601 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-23 11:35:52,704 INFO impl.YarnClientImpl: Submitted application application_1774229805073_0011
2026-03-23 11:35:52,775 INFO mapreduce.Job: The url to track the job: http://UIT.myguest.virtualbox.org:8088/proxy/application_1774229805073_0011/
2026-03-23 11:35:52,776 INFO mapreduce.Job: Running job: job_1774229805073_0011
2026-03-23 11:36:00,990 INFO mapreduce.Job: Job job_1774229805073_0011 running in uber mode : false
2026-03-23 11:36:00,993 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-23 11:36:10,270 INFO mapreduce.Job:  map 50% reduce 0%
2026-03-23 11:36:11,281 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-23 11:36:17,365 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-23 11:36:18,382 INFO mapreduce.Job: Job job_1774229805073_0011 completed successfully
2026-03-23 11:36:18,549 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=1406
                FILE: Number of bytes written=830850
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=5394
                HDFS: Number of bytes written=3050
                HDFS: Number of read operations=12
                HDFS: Number of large read operations=0
```



```
                Reduce input groups=50
                Reduce shuffle bytes=1412
                Reduce input records=100
                Reduce output records=51
                Spilled Records=200
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=258
                CPU time spent (ms)=1220
                Physical memory (bytes) snapshot=640212992
                Virtual memory (bytes) snapshot=7915323392
                Total committed heap usage (bytes)=355999744
                Peak Map Physical memory (bytes)=250617856
                Peak Map Virtual memory (bytes)=2638753792
                Peak Reduce Physical memory (bytes)=142540800
                Peak Reduce Virtual memory (bytes)=2641350656
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=2696
        File Output Format Counters
                Bytes Written=3050
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hdfs dfs -cat /output11/part-r-00000
Toy Story (1995)        AverageRating: 4.25 (TotalRatings: 2)
The Shawshank Redemption (1994) AverageRating: 4.0 (TotalRatings: 2)
Back to the Future (1985)       AverageRating: 4.0 (TotalRatings: 2)
Gladiator (2000)        AverageRating: 3.25 (TotalRatings: 2)
Braveheart (1995)       AverageRating: 4.25 (TotalRatings: 2)
Jurassic Park (1993)    AverageRating: 4.0 (TotalRatings: 2)
The Departed (2006)     AverageRating: 3.25 (TotalRatings: 2)
The Silence of the Lambs (1991) AverageRating: 3.5 (TotalRatings: 2)
Saving Private Ryan (1998)      AverageRating: 4.0 (TotalRatings: 2)
Avatar (2009)   AverageRating: 4.75 (TotalRatings: 2)
Interstellar (2014)     AverageRating: 3.5 (TotalRatings: 2)
The Prestige (2006)     AverageRating: 4.0 (TotalRatings: 2)
```


Activities
Terminal
Mar 23 11:42
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01
The Prestige (2006) AverageRating: 4.0 (TotalRatings: 2)
The Avengers (2012) AverageRating: 3.75 (TotalRatings: 2)
Memento (2000) AverageRating: 4.0 (TotalRatings: 2)
The Green Mile (1999) AverageRating: 3.5 (TotalRatings: 2)
The Matrix (1999) AverageRating: 3.75 (TotalRatings: 2)
The Lord of the Rings: The Fellowship of the Ring (2001) AverageRating: 4.25 (TotalRatings: 2)
The Lord of the Rings: The Two Towers (2002) AverageRating: 3.75 (TotalRatings: 2)
The Lord of the Rings: The Return of the King (2003) AverageRating: 3.5 (TotalRatings: 2)
Fight Club (1999) AverageRating: 4.25 (TotalRatings: 2)
Se7en (1995) AverageRating: 3.75 (TotalRatings: 2)
The Usual Suspects (1995) AverageRating: 4.25 (TotalRatings: 2)
Schindler's List (1993) AverageRating: 4.25 (TotalRatings: 2)
American Beauty (1999) AverageRating: 3.75 (TotalRatings: 2)
Good Will Hunting (1997) AverageRating: 3.5 (TotalRatings: 2)
The Social Network (2010) AverageRating: 3.75 (TotalRatings: 2)
Slumdog Millionaire (2008) AverageRating: 4.25 (TotalRatings: 2)
Whiplash (2014) AverageRating: 3.75 (TotalRatings: 2)
La La Land (2016) AverageRating: 4.0 (TotalRatings: 2)
Inception (2010) AverageRating: 3.75 (TotalRatings: 2)
Mad Max: Fury Road (2015) AverageRating: 4.0 (TotalRatings: 2)
The Revenant (2015) AverageRating: 3.5 (TotalRatings: 2)
The Dark Knight (2008) AverageRating: 3.5 (TotalRatings: 2)
Gravity (2013) AverageRating: 4.25 (TotalRatings: 2)
The Wolf of Wall Street (2013) AverageRating: 4.25 (TotalRatings: 2)
Birdman (2014) AverageRating: 3.0 (TotalRatings: 2)
Pulp Fiction (1994) AverageRating: 4.0 (TotalRatings: 2)
The Big Short (2015) AverageRating: 4.5 (TotalRatings: 2)
Manchester by the Sea (2016) AverageRating: 4.0 (TotalRatings: 2)
Moonlight (2016) AverageRating: 3.5 (TotalRatings: 2)
Spotlight (2015) AverageRating: 4.25 (TotalRatings: 2)
The Imitation Game (2014) AverageRating: 3.75 (TotalRatings: 2)
Forrest Gump (1994) AverageRating: 3.75 (TotalRatings: 2)
Dunkirk (2017) AverageRating: 3.5 (TotalRatings: 2)
Coco (2017) AverageRating: 4.25 (TotalRatings: 2)
Parasite (2019) AverageRating: 4.0 (TotalRatings: 2)
The Godfather (1972) AverageRating: 3.5 (TotalRatings: 2)
Titanic (1997) AverageRating: 3.5 (TotalRatings: 2)
Star Wars: A New Hope (1977) AverageRating: 3.5 (TotalRatings: 2)
The Lion King (1994) AverageRating: 4.0 (TotalRatings: 2)
RESULT No movie has at least 5 ratings.
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$

Câu 2:


Activities
Terminal
Mar 23 12:42
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01
Deleted /output_bai2
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hadoop jar bai2_lab1.jar Bai2_Lab1
2026-03-23 12:36:23,935 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-23 12:36:24,620 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-03-23 12:36:24,693 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/vy23521830/.staging/job_1774229805073_0013
2026-03-23 12:36:25,268 INFO input.FileInputFormat: Total input files to process : 2
2026-03-23 12:36:26,247 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-23 12:36:26,718 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774229805073_0013
2026-03-23 12:36:26,718 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-23 12:36:27,092 INFO conf.Configuration: resource-types.xml not found
2026-03-23 12:36:27,092 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-23 12:36:27,287 INFO impl.YarnClientImpl: Submitted application application_1774229805073_0013
2026-03-23 12:36:27,463 INFO mapreduce.Job: The url to track the job: http://UIT.myguest.virtualbox.org:8088/proxy/application_1774229805073_0013/
2026-03-23 12:36:27,464 INFO mapreduce.Job: Running job: job_1774229805073_0013
2026-03-23 12:36:37,895 INFO mapreduce.Job: Job job_1774229805073_0013 running in uber mode : false
2026-03-23 12:36:37,895 INFO mapreduce.Job: map 0% reduce 0%
2026-03-23 12:36:58,628 INFO mapreduce.Job: map 100% reduce 0%
2026-03-23 12:37:05,724 INFO mapreduce.Job: map 100% reduce 100%
2026-03-23 12:37:06,791 INFO mapreduce.Job: Job job_1774229805073_0013 completed successfully
2026-03-23 12:37:07,246 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=4470
FILE: Number of bytes written=836966
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=7880
HDFS: Number of bytes written=479
HDFS: Number of read operations=13
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0
Job Counters
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=35936
Total time spent by all reduces in occupied slots (ms)=4590
Total time spent by all map tasks (ms)=35936

Activities Terminal Mar 23 12:42

vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01

```
                Total time spent by all map tasks (ms)=35936
                Total time spent by all reduce tasks (ms)=4590
                Total vcore-milliseconds taken by all map tasks=35936
                Total vcore-milliseconds taken by all reduce tasks=4590
                Total megabyte-milliseconds taken by all map tasks=36798464
                Total megabyte-milliseconds taken by all reduce tasks=4700160
        Map-Reduce Framework
                Map input records=100
                Map output records=256
                Map output bytes=3952
                Map output materialized bytes=4476
                Input split bytes=212
                Combine input records=0
                Combine output records=0
                Reduce input groups=17
                Reduce shuffle bytes=4476
                Reduce input records=256
                Reduce output records=17
                Spilled Records=512
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=577
                CPU time spent (ms)=3140
                Physical memory (bytes) snapshot=640356352
                Virtual memory (bytes) snapshot=7921385472
                Total committed heap usage (bytes)=355999744
                Peak Map Physical memory (bytes)=245067776
                Peak Map Virtual memory (bytes)=2639347712
                Peak Reduce Physical memory (bytes)=151396352
                Peak Reduce Virtual memory (bytes)=2643275776
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=2696
        File Output Format Counters
```

Activities Terminal Mar 23 12:41

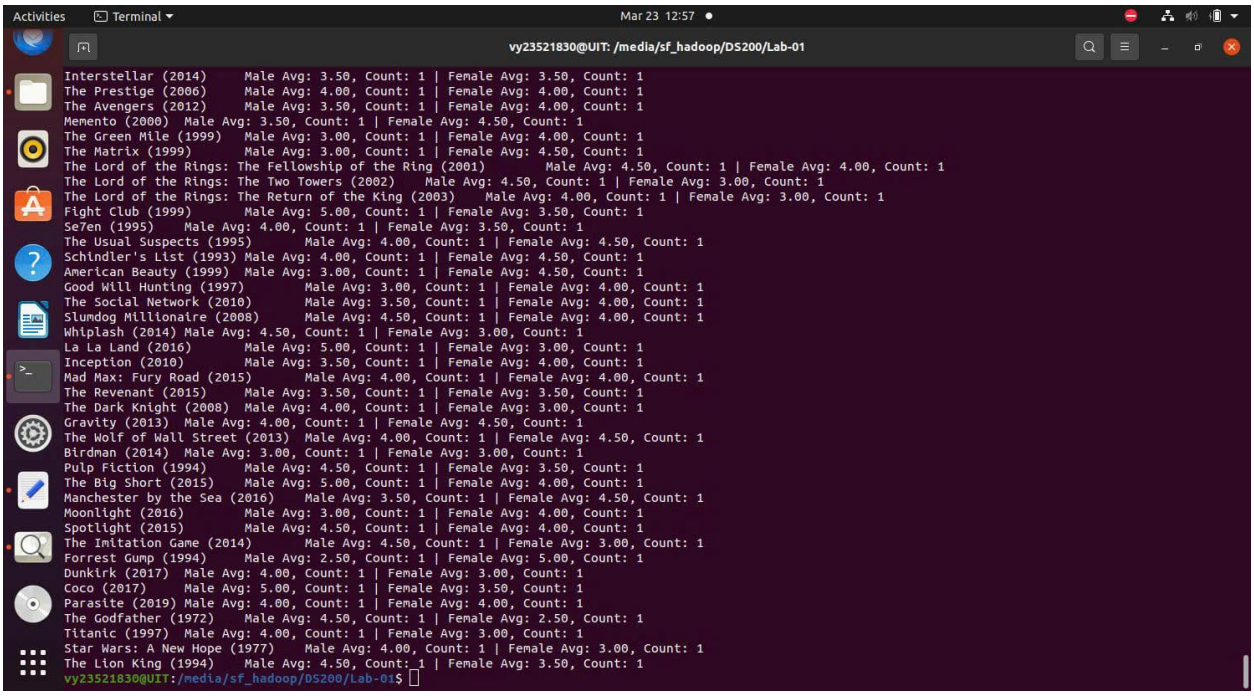
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01

```
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=577
                CPU time spent (ms)=3140
                Physical memory (bytes) snapshot=640356352
                Virtual memory (bytes) snapshot=7921385472
                Total committed heap usage (bytes)=355999744
                Peak Map Physical memory (bytes)=245067776
                Peak Map Virtual memory (bytes)=2639347712
                Peak Reduce Physical memory (bytes)=151396352
                Peak Reduce Virtual memory (bytes)=2643275776
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=2696
        File Output Format Counters
                Bytes Written=479
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hdfs dfs -cat /output_bai2/part-r-00000
Action  Avg: 3.73, Count: 20
Adventure       Avg: 3.85, Count: 30
Animation       Avg: 4.17, Count: 6
Biography       Avg: 4.14, Count: 14
Children        Avg: 4.25, Count: 2
Comedy  Avg: 3.96, Count: 12
Crime   Avg: 3.78, Count: 20
Drama   Avg: 3.81, Count: 76
Family  Avg: 4.25, Count: 2
Fantasy Avg: 3.95, Count: 10
History Avg: 4.00, Count: 6
Music   Avg: 3.88, Count: 4
Mystery Avg: 4.00, Count: 8
Romance Avg: 3.75, Count: 8
Sci-Fi  Avg: 3.85, Count: 20
Thriller        Avg: 3.88, Count: 16
War     Avg: 4.00, Count: 2
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$
```

Câu 3:

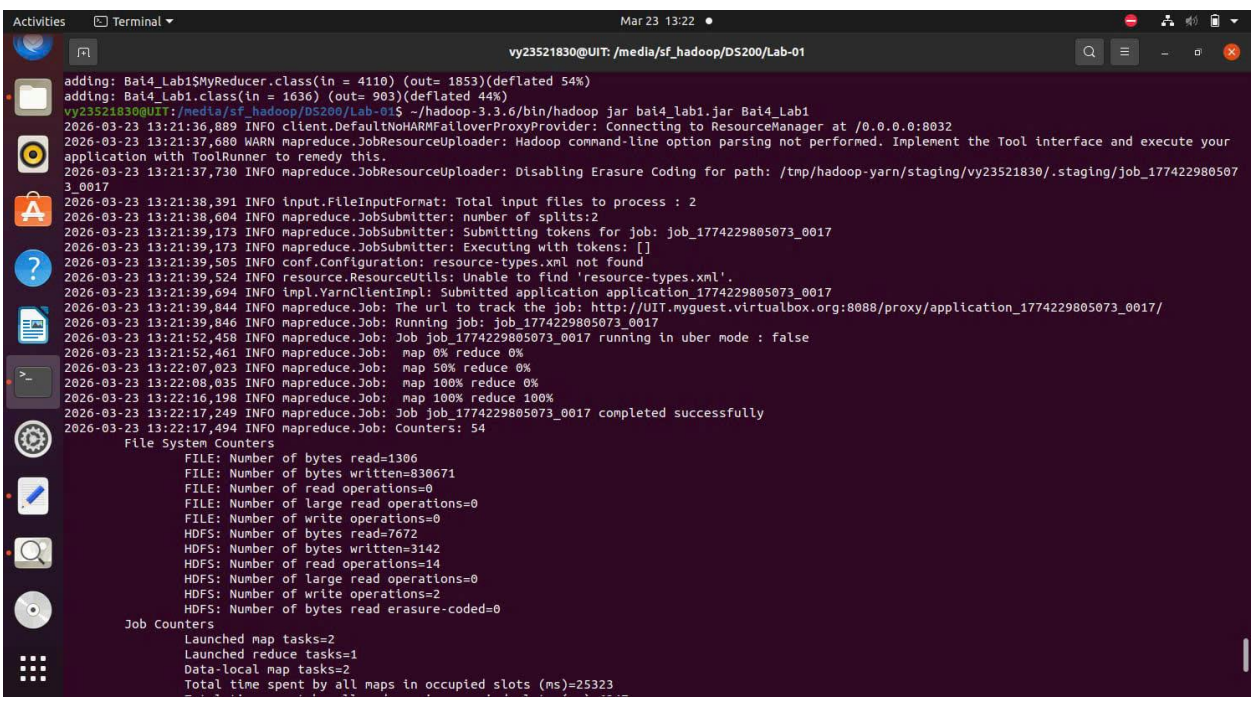
```
Activities   Terminal   Mar 23 12:57
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01
                Total vcore-milliseconds taken by all reduce tasks=3543
                Total megabyte-milliseconds taken by all map tasks=16345088
                Total megabyte-milliseconds taken by all reduce tasks=3628032
        Map-Reduce Framework
                Map input records=100
                Map output records=100
                Map output bytes=1000
                Map output materialized bytes=1212
                Input split bytes=212
                Combine input records=0
                Combine output records=0
                Reduce input groups=50
                Reduce shuffle bytes=1212
                Reduce input records=100
                Reduce output records=50
                Spilled Records=200
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=216
                CPU time spent (ms)=1150
                Physical memory (bytes) snapshot=625438720
                Virtual memory (bytes) snapshot=7918497792
                Total committed heap usage (bytes)=355999744
                Peak Map Physical memory (bytes)=242515968
                Peak Map Virtual memory (bytes)=2637418496
                Peak Reduce Physical memory (bytes)=140505088
                Peak Reduce Virtual memory (bytes)=2645430272
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=2696
        File Output Format Counters
                Bytes Written=3836
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hdfs dfs -cat /output13/part-r-00000
Toy Story (1995)        Male Avg: 4.50, Count: 1 | Female Avg: 4.00, Count: 1
```

```
Activities   Terminal   Mar 23 12:58
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hadoop jar bai3_lab1.jar Bai3_Lab1
2026-03-23 12:56:44,433 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-23 12:56:44,979 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your
application with ToolRunner to remedy this.
2026-03-23 12:56:45,025 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/vy23521830/.staging/job_177422980507
3_0015
2026-03-23 12:56:45,591 INFO input.FileInputFormat: Total input files to process : 2
2026-03-23 12:56:45,755 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-23 12:56:46,200 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774229805073_0015
2026-03-23 12:56:46,200 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-23 12:56:46,518 INFO conf.Configuration: resource-types.xml not found
2026-03-23 12:56:46,518 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-23 12:56:46,628 INFO impl.YarnClientImpl: Submitted application application_1774229805073_0015
2026-03-23 12:56:46,724 INFO mapreduce.Job: The url to track the job: http://UIT.myguest.virtualbox.org:8088/proxy/application_1774229805073_0015/
2026-03-23 12:56:46,725 INFO mapreduce.Job: Running job: job_1774229805073_0015
2026-03-23 12:56:57,100 INFO mapreduce.Job: Job job_1774229805073_0015 running in uber mode : false
2026-03-23 12:56:57,101 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-23 12:57:07,311 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-23 12:57:13,412 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-23 12:57:14,431 INFO mapreduce.Job: Job job_1774229805073_0015 completed successfully
2026-03-23 12:57:14,606 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=1206
                FILE: Number of bytes written=830465
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=7672
                HDFS: Number of bytes written=3836
                HDFS: Number of read operations=14
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=15962
                Total time spent by all reduces in occupied slots (ms)=3543
                Total time spent by all map tasks (ms)=15962
                Total time spent by all reduce tasks (ms)=3543
```



```
Interstellar (2014)     Male Avg: 3.50, Count: 1 | Female Avg: 3.50, Count: 1
The Prestige (2006)     Male Avg: 4.00, Count: 1 | Female Avg: 4.00, Count: 1
The Avengers (2012)     Male Avg: 3.50, Count: 1 | Female Avg: 4.00, Count: 1
Memento (2000)  Male Avg: 3.50, Count: 1 | Female Avg: 4.50, Count: 1
The Green Mile (1999)   Male Avg: 3.00, Count: 1 | Female Avg: 4.00, Count: 1
The Matrix (1999)       Male Avg: 3.00, Count: 1 | Female Avg: 4.50, Count: 1
The Lord of the Rings: The Fellowship of the Ring (2001)        Male Avg: 4.50, Count: 1 | Female Avg: 4.00, Count: 1
The Lord of the Rings: The Two Towers (2002)    Male Avg: 4.50, Count: 1 | Female Avg: 3.00, Count: 1
The Lord of the Rings: The Return of the King (2003)    Male Avg: 4.00, Count: 1 | Female Avg: 3.00, Count: 1
Fight Club (1999)       Male Avg: 5.00, Count: 1 | Female Avg: 3.50, Count: 1
Se7en (1995)    Male Avg: 4.00, Count: 1 | Female Avg: 3.50, Count: 1
The Usual Suspects (1995)       Male Avg: 4.00, Count: 1 | Female Avg: 4.50, Count: 1
Schindler's List (1993) Male Avg: 4.00, Count: 1 | Female Avg: 4.50, Count: 1
American Beauty (1999)  Male Avg: 3.00, Count: 1 | Female Avg: 4.50, Count: 1
Good Will Hunting (1997)        Male Avg: 3.00, Count: 1 | Female Avg: 4.00, Count: 1
The Social Network (2010)       Male Avg: 3.50, Count: 1 | Female Avg: 4.00, Count: 1
Slumdog Millionaire (2008)      Male Avg: 4.50, Count: 1 | Female Avg: 4.00, Count: 1
Whiplash (2014) Male Avg: 4.50, Count: 1 | Female Avg: 3.00, Count: 1
La La Land (2016)       Male Avg: 5.00, Count: 1 | Female Avg: 3.00, Count: 1
Inception (2010)        Male Avg: 3.50, Count: 1 | Female Avg: 4.00, Count: 1
Mad Max: Fury Road (2015)       Male Avg: 4.00, Count: 1 | Female Avg: 4.00, Count: 1
The Revenant (2015)     Male Avg: 3.50, Count: 1 | Female Avg: 3.50, Count: 1
The Dark Knight (2008)  Male Avg: 4.00, Count: 1 | Female Avg: 3.00, Count: 1
Gravity (2013)  Male Avg: 4.00, Count: 1 | Female Avg: 4.50, Count: 1
The Wolf of Wall Street (2013)  Male Avg: 4.00, Count: 1 | Female Avg: 4.50, Count: 1
Birdman (2014)  Male Avg: 3.00, Count: 1 | Female Avg: 3.00, Count: 1
Pulp Fiction (1994)     Male Avg: 4.50, Count: 1 | Female Avg: 3.50, Count: 1
The Big Short (2015)    Male Avg: 5.00, Count: 1 | Female Avg: 4.00, Count: 1
Manchester by the Sea (2016)    Male Avg: 3.50, Count: 1 | Female Avg: 4.50, Count: 1
Moonlight (2016)        Male Avg: 3.00, Count: 1 | Female Avg: 4.00, Count: 1
Spotlight (2015)        Male Avg: 4.50, Count: 1 | Female Avg: 4.00, Count: 1
The Imitation Game (2014)       Male Avg: 4.50, Count: 1 | Female Avg: 3.00, Count: 1
Forrest Gump (1994)     Male Avg: 2.50, Count: 1 | Female Avg: 5.00, Count: 1
Dunkirk (2017)  Male Avg: 4.00, Count: 1 | Female Avg: 3.00, Count: 1
Coco (2017)     Male Avg: 5.00, Count: 1 | Female Avg: 3.50, Count: 1
Parasite (2019) Male Avg: 4.00, Count: 1 | Female Avg: 4.00, Count: 1
The Godfather (1972)    Male Avg: 4.50, Count: 1 | Female Avg: 2.50, Count: 1
Titanic (1997)  Male Avg: 4.00, Count: 1 | Female Avg: 3.00, Count: 1
Star Wars: A New Hope (1977)    Male Avg: 4.00, Count: 1 | Female Avg: 3.00, Count: 1
The Lion King (1994)    Male Avg: 4.50, Count: 1 | Female Avg: 3.50, Count: 1
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$
```

Câu 4:



```
adding: Bai4_Lab1$MyReducer.class(in = 4110) (out= 1853)(deflated 54%)
adding: Bai4_Lab1.class(in = 1636) (out= 903)(deflated 44%)
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hadoop jar bai4_lab1.jar Bai4_Lab1
2026-03-23 13:21:36,889 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-23 13:21:37,680 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your
application with ToolRunner to remedy this.
2026-03-23 13:21:37,730 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/vy23521830/.staging/job_177422980507
3_0017
2026-03-23 13:21:38,391 INFO input.FileInputFormat: Total input files to process : 2
2026-03-23 13:21:38,604 INFO mapreduce.JobSubmitter: number of splits:2
2026-03-23 13:21:39,173 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774229805073_0017
2026-03-23 13:21:39,173 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-23 13:21:39,505 INFO conf.Configuration: resource-types.xml not found
2026-03-23 13:21:39,524 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-23 13:21:39,694 INFO impl.YarnClientImpl: Submitted application application_1774229805073_0017
2026-03-23 13:21:39,844 INFO mapreduce.Job: The url to track the job: http://UIT.myguest.virtualbox.org:8088/proxy/application_1774229805073_0017/
2026-03-23 13:21:39,846 INFO mapreduce.Job: Running job: job_1774229805073_0017
2026-03-23 13:21:52,458 INFO mapreduce.Job: Job job_1774229805073_0017 running in uber mode : false
2026-03-23 13:21:52,461 INFO mapreduce.Job:  map 0% reduce 0%
2026-03-23 13:22:07,023 INFO mapreduce.Job:  map 50% reduce 0%
2026-03-23 13:22:08,035 INFO mapreduce.Job:  map 100% reduce 0%
2026-03-23 13:22:16,198 INFO mapreduce.Job:  map 100% reduce 100%
2026-03-23 13:22:17,249 INFO mapreduce.Job: Job job_1774229805073_0017 completed successfully
2026-03-23 13:22:17,494 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=1306
                FILE: Number of bytes written=830671
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=7672
                HDFS: Number of bytes written=3142
                HDFS: Number of read operations=14
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=25323
```

```
Activities  Terminal  Mar 23 13:23
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01
                Total vcore-milliseconds taken by all reduce tasks=6247
                Total megabyte-milliseconds taken by all map tasks=25930752
                Total megabyte-milliseconds taken by all reduce tasks=6396928
        Map-Reduce Framework
                Map input records=100
                Map output records=100
                Map output bytes=1100
                Map output materialized bytes=1312
                Input split bytes=212
                Combine input records=0
                Combine output records=0
                Reduce input groups=50
                Reduce shuffle bytes=1312
                Reduce input records=100
                Reduce output records=50
                Spilled Records=200
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=387
                CPU time spent (ms)=2030
                Physical memory (bytes) snapshot=636743680
                Virtual memory (bytes) snapshot=7918997504
                Total committed heap usage (bytes)=355999744
                Peak Map Physical memory (bytes)=249151488
                Peak Map Virtual memory (bytes)=2638929920
                Peak Reduce Physical memory (bytes)=140816384
                Peak Reduce Virtual memory (bytes)=2641276928
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=2696
        File Output Format Counters
                Bytes Written=3142
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$ ~/hadoop-3.3.6/bin/hdfs dfs -cat /output14/part-r-00000
Toy Story (1995)        0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
```

```
Activities  Terminal  Mar 23 13:23
vy23521830@UIT: /media/sf_hadoop/DS200/Lab-01
Interstellar (2014)     0-18: NA 18-35: 3.50 35-50: 3.50 50+: NA
The Prestige (2006)     0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
The Avengers (2012)     0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Memento (2000)  0-18: NA 18-35: 4.50 35-50: 3.50 50+: NA
The Green Mile (1999)   0-18: NA 18-35: 4.00 35-50: 3.00 50+: NA
The Matrix (1999)       0-18: NA 18-35: 3.75 35-50: NA 50+: NA
The Lord of the Rings: The Fellowship of the Ring (2001)        0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
The Lord of the Rings: The Two Towers (2002)    0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
The Lord of the Rings: The Return of the King (2003)    0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Fight Club (1999)       0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Se7en (1995)    0-18: NA 18-35: 3.75 35-50: NA 50+: NA
The Usual Suspects (1995)       0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Schindler's List (1993) 0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
American Beauty (1999)  0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Good Will Hunting (1997)        0-18: NA 18-35: 3.50 35-50: NA 50+: NA
The Social Network (2010)       0-18: NA 18-35: 4.00 35-50: 3.50 50+: NA
Slumdog Millionaire (2008)      0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
Whiplash (2014) 0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
La La Land (2016)       0-18: NA 18-35: 3.00 35-50: 5.00 50+: NA
Inception (2010)        0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Mad Max: Fury Road (2015)       0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Revenant (2015)     0-18: NA 18-35: 3.50 35-50: NA 50+: NA
The Dark Knight (2008)  0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
Gravity (2013)  0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
The Wolf of Wall Street (2013)  0-18: NA 18-35: 4.50 35-50: 4.00 50+: NA
Birdman (2014)  0-18: NA 18-35: 3.00 35-50: NA 50+: NA
Pulp Fiction (1994)     0-18: NA 18-35: 4.00 35-50: NA 50+: NA
The Big Short (2015)    0-18: NA 18-35: 4.50 35-50: NA 50+: NA
Manchester by the Sea (2016)    0-18: NA 18-35: 4.50 35-50: 3.50 50+: NA
Moonlight (2016)        0-18: NA 18-35: 4.00 35-50: 3.00 50+: NA
Spotlight (2015)        0-18: NA 18-35: 4.00 35-50: 4.50 50+: NA
The Imitation Game (2014)       0-18: NA 18-35: 3.00 35-50: 4.50 50+: NA
Forrest Gump (1994)     0-18: NA 18-35: 3.75 35-50: NA 50+: NA
Dunkirk (2017)  0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Coco (2017)     0-18: NA 18-35: 4.25 35-50: NA 50+: NA
Parasite (2019) 0-18: NA 18-35: 4.00 35-50: 4.00 50+: NA
The Godfather (1972)    0-18: NA 18-35: 3.50 35-50: NA 50+: NA
Titanic (1997)  0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
Star Wars: A New Hope (1977)    0-18: NA 18-35: 3.00 35-50: 4.00 50+: NA
The Lion King (1994)    0-18: NA 18-35: 4.00 35-50: NA 50+: NA
vy23521830@UIT:/media/sf_hadoop/DS200/Lab-01$
```